REVENUELEAK

Revenue, Customer and Product Performance Analysis

Business Analysis Report

Executive Summary

This report analyzes the Online Retail II transaction dataset to identify revenue patterns, cancellation exposure, customer engagement opportunities and product combinations that may support cross-selling.

The cleaned dataset contains **1,027,017** transaction-line records across **43** countries. The analysis identifies **20,476,260.45** in sales value and **1,462,424.18** in identified cancellation value, resulting in preliminary Net Gross Revenue of **19,013,836.27**.

Business metric	Result
Total sales value	20,476,260.45
Cancellation value	1,462,424.18
Preliminary Net Gross Revenue	19,013,836.27
Sales invoices	40,077
Average Order Value	510.92
Customers in RFM analysis	5,878

Headline Findings

Market concentration: The United Kingdom contributes 84.9% of preliminary Net Gross Revenue.

Customer engagement: 2,351 customers (40.0%) are classified as At Risk or Hibernating under the rule-based RFM segmentation.

Product association: The strongest discovered rule by lift has a lift of 48.28 and confidence of 73.4%.

Purpose: Provide practical, evidence-based actions for revenue monitoring, customer retention and product recommendations.

1. Dataset and Methodology

The analysis uses the Online Retail II dataset, which contains retail transaction records from December 2009 to December 2011. Each row represents a product line within an invoice.

Data preparation approach

Area	Treatment
Duplicate records	Exact duplicate rows removed.
Invoice dates	Parsed into date-time values; invalid dates excluded.
Price	Rows with zero or negative prices excluded.
Cancellations	Invoices beginning with C handled separately.
Missing Customer IDs	Retained in revenue totals; excluded from RFM.
Negative quantities	Non-cancellation negative adjustments with non-positive prices excluded.
Storage	Cleaned records stored in Parquet and queried using DuckDB.

Revenue definitions

Sales Value is calculated from eligible positive-quantity sales lines. Cancellation Value is the absolute value of identified cancellation invoice lines. Preliminary Net Gross Revenue is calculated as Sales Value minus Cancellation Value.

Important: This dataset does not provide product cost or operating expense data. Therefore, the report does not calculate profit or profit margin. The term Net Gross Revenue is used only for sales less identified cancellation value.

The cleaned dataset contains 1,027,017 rows. Revenue from transactions with missing Customer IDs is 2,723,844.98, approximately 14.3% of preliminary Net Gross Revenue. These transactions remain included in overall revenue calculations.

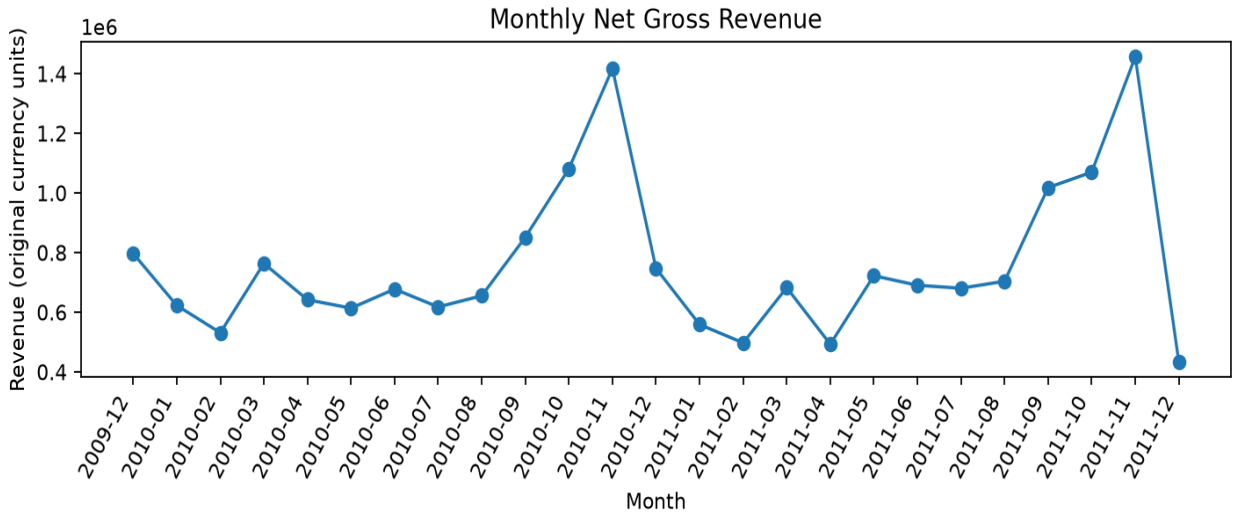
Analytical methods

Revenue trends are calculated monthly and by country. Average Order Value is the average of invoice-level positive sales values. Customer analysis uses Recency, Frequency and Monetary (RFM) measures. Market Basket Analysis uses FP-Growth to identify frequently co-purchased products and association rules.

For market basket analysis, the 500 most frequently purchased products were selected, with a minimum occurrence threshold of 20 invoices. The analysis included 37,083 invoices containing at least one selected product. Minimum support was 1%, minimum confidence was 20%, and only rules with lift greater than 1 were retained.

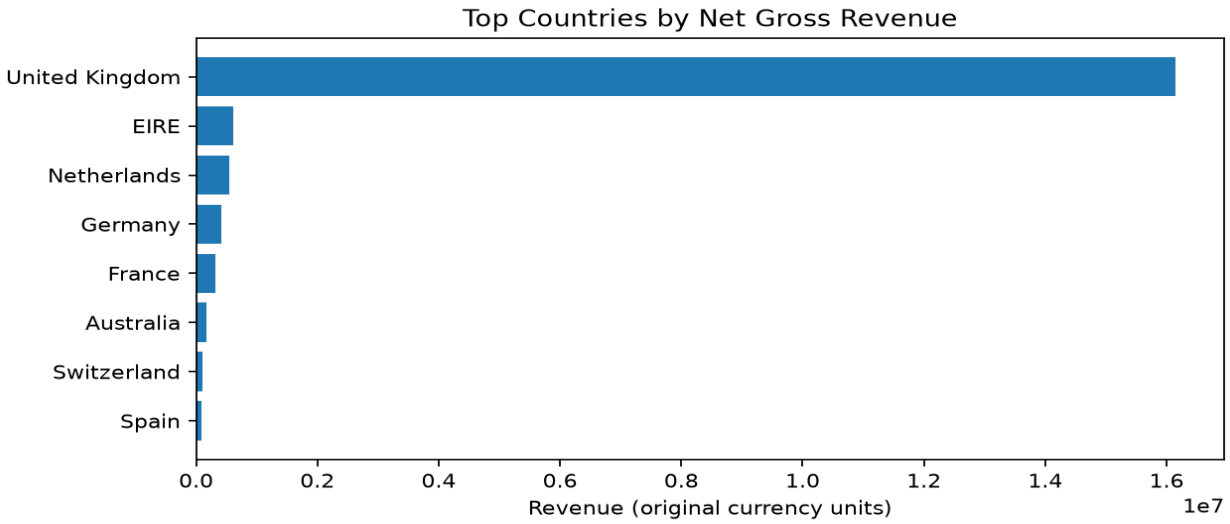
2. Revenue Performance

The monthly revenue trend provides a view of changes in transaction performance over the dataset period. Peaks and declines can help management identify periods for further operational investigation.



Country contribution

Country	Sales	Cancellation value	Net gross revenue
United Kingdom	17,410,196.12	1,265,833.21	16,144,362.91
EIRE	658,767.31	48,813.53	609,953.78
Netherlands	554,038.09	5,707.39	548,330.70
Germany	425,019.71	13,060.55	411,959.16
France	350,456.09	28,722.70	321,733.39
Australia	169,283.46	2,839.04	166,444.42
Switzerland	100,685.59	1,282.53	99,403.06
Spain	108,332.49	17,319.05	91,013.44

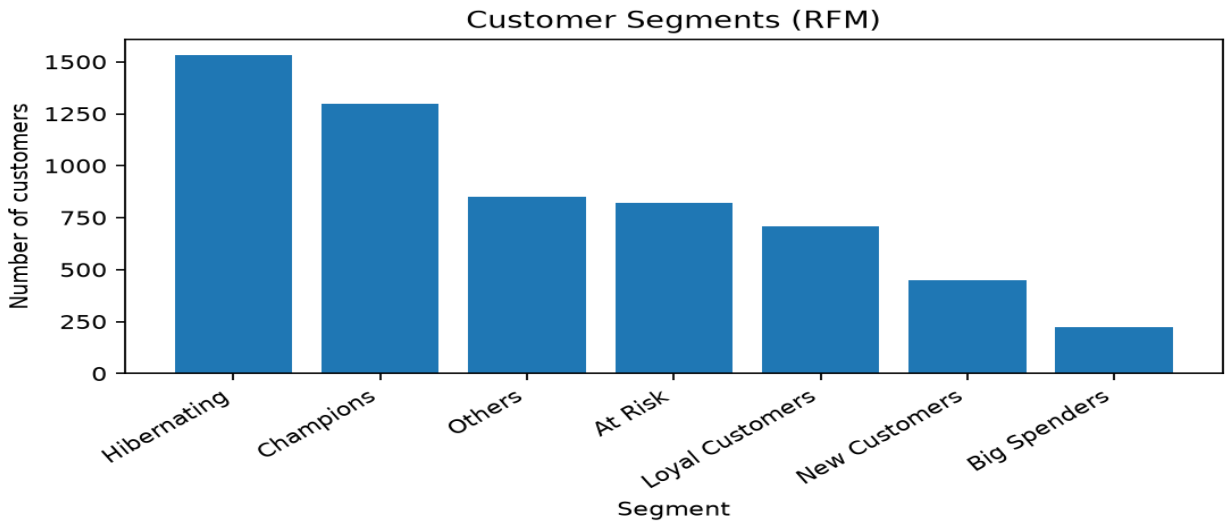


The United Kingdom is the largest market, contributing 84.9% of preliminary Net Gross Revenue (16,144,362.91). This concentration makes UK performance particularly important to overall revenue monitoring.

Average Order Value across 40,077 eligible sales invoices is 510.92. AOV is an invoice-level average of positive sales value and does not represent profit.

3. Customer and Cancellation Analysis

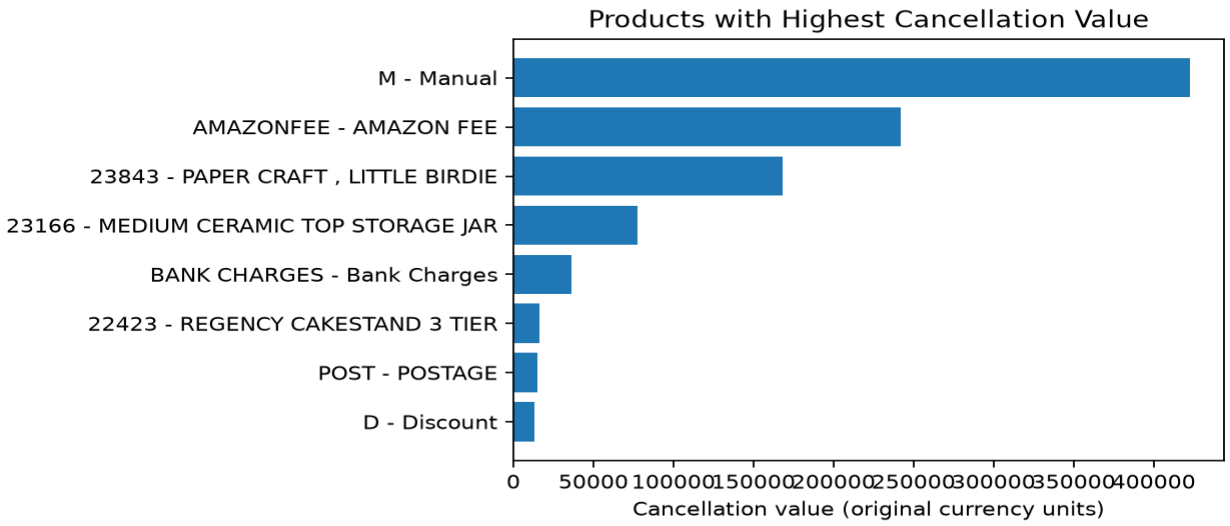
RFM analysis groups customers according to how recently they purchased, how frequently they purchased and how much positive sales value they generated. The segments are rule-based classifications, not predictive models.



RFM segment	Customers
Hibernating	1,531
Champions	1,299
Others	851
At Risk	820
Loyal Customers	707
New Customers	448
Big Spenders	222

There are 2,351 At Risk and Hibernating customers, representing 40.0% of the analyzed customer base. This group is a potential audience for carefully tested re-engagement campaigns.

Cancellation exposure by product



Product	Cancellation value
Manual	422,926.14
AMAZON FEE	241,988.30
PAPER CRAFT , LITTLE BIRDIE	168,469.60
MEDIUM CERAMIC TOP STORAGE JAR	77,479.64
Bank Charges	36,001.49

The products shown are ranked by cancellation value in cancellation invoices. These records do not necessarily identify the original sale invoice or explain why a cancellation occurred. The results should be used to prioritize investigation rather than infer product defects.

4. Product Association Analysis

Market Basket Analysis identifies products that frequently appear together in the same invoice. Support measures how common a combination is, confidence measures how often the consequent product appears when the antecedent is present, and lift compares that likelihood with the consequent product's overall frequency.

Products purchased together	Confidence	Lift
22745: POPPY'S PLAYHOUSE BEDROOM | 22748: POPPY'S PLAYHOUSE KITCHEN **Also:** 22746: POPPY'S PLAYHOUSE LIVINGROOM	73.4%	48.28
22746: POPPY'S PLAYHOUSE LIVINGROOM **Also:** 22745: POPPY'S PLAYHOUSE BEDROOM | 22748: POPPY'S PLAYH	72.5%	48.28
22746: POPPY'S PLAYHOUSE LIVINGROOM | 22748: POPPY'S PLAYHOUSE KITCHEN **Also:** 22745: POPPY'S PLAYHOUSE BEDROOM	86.3%	45.52
22745: POPPY'S PLAYHOUSE BEDROOM **Also:** 22746: POPPY'S PLAYHOUSE LIVINGROOM | 22748: POPPY'S PL	58.2%	45.52
22745: POPPY'S PLAYHOUSE BEDROOM | 22746: POPPY'S PLAYHOUSE LIVINGROOM **Also:** 22748: POPPY'S PLAYHOUSE KITCHEN	88.7%	44.34
22748: POPPY'S PLAYHOUSE KITCHEN **Also:** 22745: POPPY'S PLAYHOUSE BEDROOM | 22746: POPPY'S PLAYH	55.1%	44.34

The strongest associations include the Poppy's Playhouse Bedroom, Kitchen and Livingroom products, as well as matching mug and cup collections. These combinations may be useful for testing product bundles and recommendation placements.

5. Prioritized Business Recommendations

1. Improve customer retention. Start with At Risk and Hibernating customers. Test targeted reminders, relevant product suggestions and limited promotions. Track repeat purchase rate and revenue generated by each campaign.

2. Test cross-selling and product bundles. Use the strongest product associations to create bundle suggestions or complementary product placements. Measure conversion, order value and incremental revenue before scaling.

3. Investigate cancellations and customer identification. Prioritize the UK because it has the largest absolute cancellation value. Review cancellation records and operational processes. Improve Customer ID capture to strengthen customer-level reporting.

6. Limitations and Next Steps

The analysis is descriptive and does not establish causal relationships. Cancellation invoices may not be directly matched to original sales invoices. Cancellation-to-sales value ratios are not the percentage of invoices cancelled or a confirmed customer return rate.

RFM Monetary values are based on positive Sales Value, rather than customer-level cancellation-adjusted revenue. Market basket results are limited to the selected 500 products and the invoices containing at least one of those products. Association does not imply causation.

Recommended next steps include validating cancellation reasons, testing retention campaigns, evaluating bundle performance and improving customer identification in transaction records.

Conclusion: RevenueLeak provides a consolidated view of revenue, customer engagement, cancellation exposure and product associations. The findings provide a practical starting point for focused business experiments and ongoing performance monitoring.